

9. class, Electrodynamics and Optics

Page 826, Task 31.1.

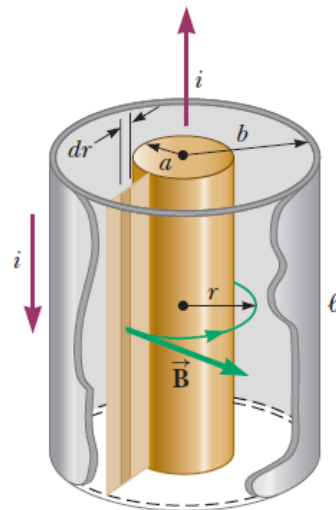
Consider a uniformly wound solenoid having N turns and length l . Assume l is much longer than the radius of the windings and the core of the solenoid is air.

- (A) Find the inductance of the solenoid.
- (B) Calculate the inductance of the solenoid if it contains 300 turns, its length is 25.0 cm, and its cross-sectional area is 4.00 cm^2 .
- (C) Calculate the self-induced emf in the solenoid if the current it carries decreases at the rate of 50.0 A/s.

Page 832, Task 31.4.

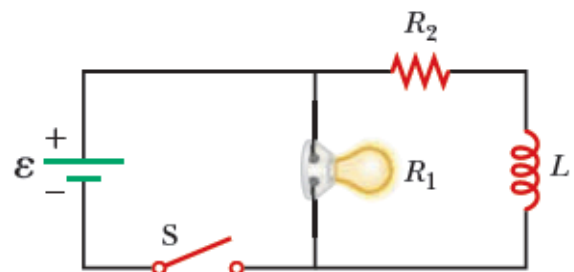
Coaxial cables are often used to connect electrical devices, such as your video system, and in receiving signals in television cable systems. Model a long coaxial cable as a thin, cylindrical conducting shell of radius b concentric with a solid cylinder of radius a as in the figure. The conductors carry the same current i in opposite directions.

Calculate the inductance L of a length l of this cable.



Page 842, Task 14.

You are working as a demonstration assistant for a physics professor. He shows you the circuit in the figure, which he wants you to build for an upcoming class. The lightbulb is a household incandescent bulb that receives energy at the rate of 40.0 W when operating at 120 V. It has a resistance R_1 , which, for simplicity, we will assume is constant at all operating voltages. The battery in the circuit has an emf of 12.0 V. When the switch has been closed for a long time, the bulb glows dimly, since it is powered by only 12.0 V. When the switch is opened, however, the bulb flashes brightly and then gradually dims to darkness. Your professor wants you to determine two values:



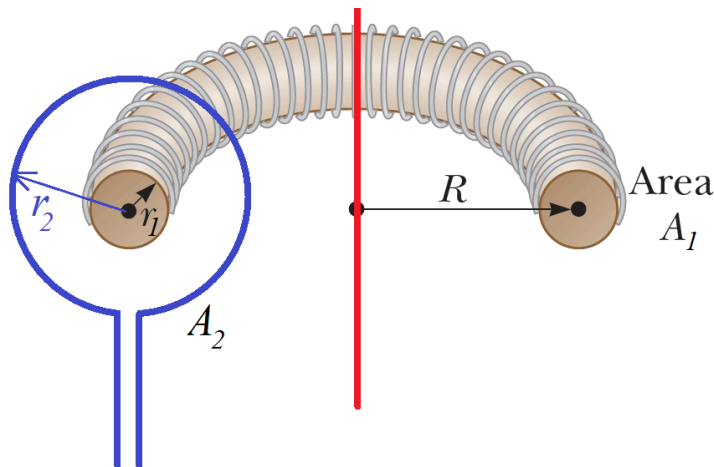
- (a) the resistance R_2 that is necessary for the bulb to initially flash, when the switch is opened, at the same brightness it would have if plugged into a 120-V socket;
- (b) the inductance L necessary to keep the current in the lightbulb above 50.0% of its value when the switch is opened, for a time interval of 2.00 s after it is opened. Assume a resistance-free inductor and that the resistance of the lightbulb does not vary with temperature.

Based on Page 841, Task 6.

A toroid has a major radius R and a minor radius r_1 and is tightly wound with N turns of wire on a hollow cardboard torus. The figure shows half of this toroid, allowing us to see its cross section. If $R \gg r_1$, the magnetic field in the region enclosed by the wire is essentially the same as the magnetic field of a solenoid that has been bent into a large circle of radius R .

Modeling the field as the uniform field of a long solenoid, calculate the mutual inductance

- (a) between the red straight wire and the toroid,
- (b) between the blue loop and the toroid.



Page 842, Task 23.

A 24.0-V battery is connected in series with a resistor and an inductor, with $R = 8.00 \, \Omega$ and $L = 4.00 \, \text{H}$, respectively. Find the energy stored in the inductor

- (a) when the current reaches its maximum value
- (b) at an instant that is a time interval of one time constant after the switch is closed.

Page 842, Task 24.

A flat coil of wire has an inductance of 40.0 mH and a resistance of 5.00 Ω . It is connected to a 22.0-V battery at the instant $t = 0$. Consider the moment when the current is 3.00 A.

- (a) At what rate is energy being delivered by the battery?
- (b) What is the power being delivered to the resistance of the coil?
- (c) At what rate is energy being stored in the magnetic field of the coil?
- (d) What is the relationship among these three power values?
- (e) Is the relationship described in part (d) true at other instants as well?
- (f) Explain the relationship at the moment immediately after $t = 0$ and at a moment several seconds later.